

AKSHAY DEEP SIDDAM

akshaysiddam.baruch@gmail.com | +1 551 394 1585 | [LinkedIn](#)

PROFESSIONAL SUMMARY

Data Engineer with 5+ years of extensive experience in building scalable **cloud data platforms, real-time streaming systems, and AI-powered analytics solutions** across banking and enterprise domains. Expertise in **AWS and Azure ecosystems**, including Databricks, Glue, EMR, Synapse, S3, and Kinesis, handling large-scale enterprise data with high performance and reliability. Strong hands-on experience in **Generative AI, LLMs, RAG pipelines, and vector databases (OpenSearch, Pinecone, FAISS)** to deliver intelligent search and customer intelligence platforms with improved data retrieval accuracy. Skilled in **PySpark, Spark, Airflow, Kafka, and ETL optimization**, enhancing pipeline efficiency along with strong expertise in **CI/CD automation, Terraform, Docker, and Kubernetes (EKS)** for scalable deployments. Proficient in **data governance, security, and compliance (IAM, RBAC, KMS, Lake Formation)** and delivering enterprise-grade BI solutions using **Power BI and Tableau** for data-driven decision-making.

TECHNICAL SKILLS

Cloud Platforms: AWS, Azure

Big Data & Data Processing: Apache Spark, PySpark, Databricks, Apache Airflow, Hadoop

Data Storage & Warehousing: Amazon S3, AWS Glue Data Catalog, Amazon Redshift, Azure Data Lake Storage (ADLS Gen2), Azure Synapse Analytics

ETL / Data Integration: AWS Glue, Azure Data Factory, ETL/ELT Pipelines

Data Transformation & Modeling: dbt, Dimensional Modeling, Star/Snowflake Schema, Data Normalization

Streaming & Real-Time Data: Apache Kafka, Amazon Kinesis, Azure Event Hubs, Azure Stream Analytics, AWS Lambda

Programming Languages: Python, SQL

AI / ML & GenAI: Generative AI, LLMs, RAG, LangChain, Amazon Bedrock, Azure OpenAI

Vector Databases & Search: OpenSearch, Pinecone, FAISS

CI/CD & DevOps: AWS CodePipeline, AWS CodeBuild, Azure DevOps, Jenkins, GitHub Actions

Infrastructure & Containerization: Docker, Kubernetes (Amazon EKS), Terraform

Data Governance & Security: IAM, RBAC, AWS KMS, AWS Lake Formation, Azure Key Vault, Managed Identities

BI & Visualization: Power BI, Tableau

PROFESSIONAL EXPERIENCE

Amazon, USA | Data Engineer

May 2025 – Present

Project: Enterprise AI-Powered Customer Intelligence & Search Platform

Description: Designed and implemented a scalable AI-driven data platform enabling business teams to analyze customer behavior, product catalog data, operational metrics, and support interactions using Generative AI, Large Language Models (LLMs), and Retrieval-Augmented Generation (RAG). Leveraged AWS cloud services to build secure, high-performance data pipelines, semantic search capabilities, and real-time analytics solutions across enterprise datasets.

Responsibilities:

- Built and optimized end-to-end RAG pipelines using Amazon Bedrock, LangChain, and OpenSearch/Pinecone, improving enterprise search relevance and retrieval accuracy by 40–45% across customer and product datasets.
- Developed scalable ETL/ELT pipelines using AWS Glue, EMR, PySpark, S3, and dbt, writing complex SQL transformations and modeling dimensional schemas (fact/dimension tables) to process 8–10M+ records daily from multi-domain sources including customer behavior, catalog, and support data.
- Designed and implemented semantic search and embedding generation pipelines using Amazon Bedrock embeddings and vector search, reducing average query latency by 30–35%.
- Built real-time streaming data pipelines using Amazon Kinesis, Kafka, and AWS Lambda, enabling near real-time analytics with data freshness improved by 60%.
- Architected and optimized data lake and warehouse solutions using Amazon S3, Glue Data Catalog, and Redshift, enabling unified analytics and reducing reporting data preparation time by 40%.
- Tuned and optimized Spark workloads on AWS EMR, improving job performance by 45–50% through partitioning, caching strategies, and cluster optimization.
- Containerized and deployed AI/data services using Docker and Amazon EKS, improving deployment scalability and reducing environment setup time by 35%.

- Implemented secure data governance, automated data validation using AWS Deequ, and CI/CD automation using AWS IAM, Lake Formation, KMS, CodePipeline, and CodeBuild, reducing deployment time by 30% and cloud compute costs by 20% while ensuring enterprise-grade security compliance.

Wells Fargo, India | Data Engineer

Jun 2020 – Jul 2024

Project: Enterprise Cloud Data Lake Modernization

Description: Led modernization of legacy banking data systems into a scalable Azure cloud-based data lake and analytics platform, enabling batch and real-time financial data processing for risk management, fraud detection, and regulatory reporting systems.

Responsibilities:

- Designed and developed end-to-end Azure data pipelines using Azure Data Factory, Azure Databricks, ADLS Gen2, and Azure Synapse Analytics, migrating over 5+ TB of legacy data monthly into cloud environments.
- Built real-time streaming data pipelines using Azure Event Hubs, Stream Analytics, and Kafka, improving fraud detection and event processing latency by 40%.
- Optimized large-scale ETL workflows using PySpark, Azure Databricks, and Apache Airflow, improving overall data processing efficiency and job performance by 55%.
- Developed and maintained dimensional data models (star schema with fact and dimension tables) using SQL for credit risk, AML, and transaction analytics, supporting regulatory reporting and business intelligence use cases.
- Implemented data quality and validation frameworks using dbt tests and Great Expectations, including schema checks, anomaly detection, and reconciliation rules, reducing data inconsistencies by 30%.
- Automated CI/CD and infrastructure deployment using Terraform, Azure DevOps, and Jenkins, reducing release and deployment cycles by 60%.
- Integrated enterprise BI solutions using Power BI, delivering interactive dashboards for executive reporting and financial risk analysis.
- Enforced enterprise security and compliance using Azure RBAC, Key Vault, Managed Identities, and PCI-DSS policies, ensuring secure and compliant data access across systems.
- Supported on-premise Oracle to Azure cloud migration, enabling scalable cloud-native architecture and improving system reliability and data availability.

ACADEMIC PROJECTS

AI-Powered Customer Insights Platform (RAG-Based Analytics System)

Designed and developed an AI-driven analytics platform using Python, Azure Databricks, and Azure OpenAI, enabling natural language querying over structured and unstructured datasets. Implemented RAG architecture with vector database (FAISS) to retrieve relevant business insights from large-scale datasets. Built ETL pipelines using PySpark and Azure Data Factory to process over 1M+ records and generate real-time customer behavior insights with 40% improved query accuracy.

Cloud Data Lake & Real-Time Streaming Analytics System

Built a scalable AWS-based data lake architecture using S3, Glue, and Redshift for batch and streaming data processing. Integrated Kafka and AWS Kinesis for real-time data ingestion and transformation using PySpark and Airflow orchestration. Developed optimized data models for analytics dashboards, reducing data processing time by 45% and enabling near real-time reporting for business decision-making.

EDUCATION:

Master of Business Administration (STEM) – Data Analytics | Zicklin School of Business, Baruch College, CUNY | May 2026